SAM VOLUME I

Retention, accumulation, and the finite ledger

Chapter II

Abstract. A retained account can outlast its present address. Original Write, interacting-clock, spin and outgoing-field results distinguish predictive state, timed history and energetic complement. This development adds source-linked spectral predictions, compact memory and monitor derivations, exact finite-source nonclosure certificates, a worked paid-reference protocol, and an eight-time ensemble measurement with explicit apparatus and sampling budgets. Signed phases, full source energy and physical calibration remain in the same account. A companion source-to-field extension separates complete energy from a diagnostic minimum, gives exact spatial receiving constructions, and derives an operational coupling-response criterion for the actual interacting clock.

1. The physical proposition

SAM begins with matter displacing substrate and the displaced account being retained. The aquarium thought experiment makes this relation visible: an object appearing in a filled enclosure changes the surrounding arrangement. Its physical content is the requirement to account for the consequence and its later reception. A fluid equation is an additional model choice. [1]

The recent work lets us ask this more precisely. What must a Write leave behind so that its consequence remains available to a later interaction? A present reading, a persistent source state, an ordered history, and an outgoing field can retain different parts of that account.

This chapter uses an operational distinction: two histories remain distinguishable when a specified permitted continuation gives different receiver outputs. This defines what a receiving description must preserve. It does not assign a measured energy or a spatial density to information alone.

The finite ledger organizes the available types. Its physical payload is developed through the source, evolution, reference and receiver that consume it.

2. Three meanings of a retained record

A record can be a current address, a state sufficient for future prediction, or an accumulated history statistic. These meanings already separate in native execution. One retained J4 source has eight immediate outputs but sixteen predictive states. Requiring the next Write output refines eight classes to fourteen; another step gives sixteen. Every distinct pair has a separating word of length at most two. [3]

The same source supplies two 324-Write histories with the same endpoint, eight-bin occupancy, and scalar logarithmic summaries. Their first-half receiver values differ: approximately 0.1440964305 and 0.4063809938 in the declared X direction. Reordering closed blocks changes when a response occurs while preserving its whole-history occupancy. [3]

The eight immediate outputs in this assay and the eight partition addresses introduced later have separate source definitions. Their equal cardinalities supply no transition map.

Chapter I contributes another exact distinction. Its four equal-occupation source eigenstates have zero D and C currents while carrying different energies. A current record therefore leaves an energetic complement. A complete energy receiver must retain that complement even when the current is zero. [2]

The task is consequently to specify the record needed by the chosen future receiver, then carry its physical account through an actual evolution.

3. Predictive equivalence and retention

For a fixed source, event alphabet and receiver, let $O(h;u)$ denote the output obtained by extending history h with an allowed future word u. Define

$$h \sim h' \quad \Longleftrightarrow \quad O(h;u) = O(h';u) \text{ for all } u.$$

This relation is preserved by appending the same event: each subsequent test is already a permitted continuation. A quotient state can therefore preserve every future output when its classes and successor transitions obey this relation. The completed sixteen-state assay gives an explicit finite realization. [3]

A clocked record requires more. Suppose an admitted event from q has positive actual phase increment $f(q,e)$. Preserving its entire future phase trace requires preservation of the present output, every $f(q,e)$, and the successor class. Repeated partition refinement gives the coarsest such quotient in a finite instrument. This is a new receiving theorem; its phase increments are supplied source operands.

A two-state construction illustrates the distinction. Both states display one address and alternate under the sole event. Their successive phase increments are $1,2$ and $2,1$. Both have phase 3 after two events, yet their first recorded phases differ. An untimed quotient has one class; its phase-compatible quotient has two.

4. A clock-weighted history monitor

Chapter I supplies actual positive rates on a stated preparation and time interval. With continuous lifted phase θ and rate ν,

$$d\theta = \nu(\tau)d\tau, \qquad 0 < \ell \le \nu \le u.$$

For a piecewise retained label j, define its elapsed exposure H, phase exposure P, and first phase moment M:

$$H_j = \int 1_j \, d\tau, \qquad P_j = \int 1_j \, d\theta,$$
$$M_j = \int \theta 1_j \, d\theta.$$

A constant-label segment with actual phase endpoints α, β updates P by $\beta - \alpha$ and M by $(\beta^2 - \alpha^2)/2$. Prescribed phase gates use the intersection of their phase intervals with this segment. No constant within-segment rate is needed. The achieved rate bounds give

$$\ell H_j \le P_j \le u H_j.$$

For a receiver affine in phase, these counters determine its full integral exactly. Event counts, elapsed exposure and phase exposure remain separate measures. The counter family is finite for the specified receivers; its numbers can grow with the history.

The completed two-past-coordinate covariance recipient also survives a matched-node change to actual phase coordinates. Its original physical lags remain attached. Substituting raw nonlinear phase differences for those lags changes the receiver. [4]

5. Original Write action and occupation

Chapter I retains lifted coordinates

$$\gamma = \varphi_a + \varphi_b - 2\varphi_p,$$
$$\delta = \varphi_a - \varphi_b, \qquad \chi = \varphi_p,$$
$$\omega_W = \hbar(d\gamma + 3d\chi).$$

Its full pair source has swap S, occupation projector N, and even operator K_e:

$$N = (I+S)/2, \qquad H_p = 2N + K_e.$$

K_e commutes with N and both currents. It is distinct from the positive interacting field form also called K in Chapter I. [2]

For a fixed normalized source preparation ψ and n equal to its N expectation, declare the receiving lift

$$\Psi_W = e^{-3i\chi} e^{-i\gamma N} \psi.$$

Its canonical one-form is exactly

$$i\hbar \langle \Psi_W | d\Psi_W \rangle = \hbar(n\, d\gamma + 3d\chi).$$

The original one-form is recovered on the occupied class $N\psi = \psi$. The original δ coordinate and ordered word remain in the retained record; this two-coordinate receiving lift supplies no independent δ generator. It also retains the distinction between the successful current-observable map and the literal endpoint Hilbert action. [5]

6. The full even-energy payment

The full energy lift at the same current angle is

$$\Psi_H = e^{-i\gamma H_p/2} \psi.$$

Because N and K_e commute, its relative operator is

$$\Psi_W = e^{-i(3\chi I - \gamma K_e/2)} \Psi_H.$$

On a sharp K_e sector the relative factor is a scalar phase. Across coherent sectors it carries their distinct phases. For $\bar{k}$ equal to the mean even energy and V its variance, the squared generator residual is

$$\|R\|^2 = \frac{\dot{\gamma}^2}{4} V + \left(\frac{\dot{\gamma}}{2} \bar{k} - 3\dot{\chi} \right)^2.$$

This separates a mean-phase payment from an operator complement. Scalar canonical one-form matching is a weaker, separately defined question:

$$3d\chi = (n - 1 + \bar{k}/2) d\gamma.$$

For the actual placement vector, $n = 1$, $\bar{k} = 573047$, and $V = 14384109824$. Thus the occupied canonical lift is exact while the full source-energy lift retains a nonzero complement.

Original legal role tangents fix the γ/χ ratio. Their direct identification with unshifted $H_p/2$ requires an additional reference or channel payment. These explicit conditions locate the remaining physical incidence; selecting χ afterward would change the original path.

7. The actual clock preparation

In a driven lane $N = (I + \sigma_x)/2$ and $Z = \sigma_z + i\sigma_y$. Positivity of every reduced source state gives

$$|\langle Z \rangle|^2 \le 4n(1-n).$$

A nonzero clock amplitude therefore requires $0 < n < 1$. The occupied class in Section 5 has zero clock amplitude and remains a separately labelled example.

The completed hard preparation instead coherently combines exact charge-zero and charge-one branches. Its charge expectation is $1/2$ and variance $1/4$. On their charge orbit,

$$z(\gamma) = M e^{-i\gamma}, \qquad a = \frac{\hbar}{2} d\gamma.$$

The transition phase reads a charge difference; its canonical action reads the prepared charge expectation. For actual hard branch energies E_0 and E_1, their gap Ω gives $z = M \exp(-i\Omega\tau)$, while the canonical energy payment includes $E_0 + \Omega/2$. [2]

The full interacting clock retains both field polarizations and its actual joint correlator. In a general changing phase frame its connection is

$$a = \hbar(n\, d\gamma + 3d\chi + a_\eta).$$

The internal joint-state Berry form a_η carries changing preparation and entanglement. For finite-energy continuum evolution its canonical energy identity is understood through the form-domain overlap derivative.

8. A worked actual phase history

Use the completed lane01 hard-coherent/soft-vacuum preparation and its actual phase lift $\theta(0) = 0$. Its outward rate bounds, per model time, are

$$3.490102929541 \times 10^{-6} \le \nu,$$
$$\nu \le 3.490102934505 \times 10^{-6}.$$

The native certificate retains sharper exact rational endpoints. Throughout $T_0 = 573049\pi$, θ is increasing. The first three quarter-phase records occur uniquely at $\theta = k\pi/2$, $k = 1,2,3$, with

$$\frac{k\pi}{2u} \le \tau_k \le \frac{k\pi}{2\ell}.$$

Existence follows from $573049\ell > 3/2$. The fourth record is outside this certified window because $573049u < 2$. The bare cycle therefore retains three completed quarter crossings and a partial fourth; it is not a completed actual phase turn.

Each record carries k, its wrapped quarter address, lifted θ, and elapsed-time enclosure. For the conserved total joint energy E_{tot}, its dynamical energy action is $\hbar E_{\mathrm{tot}}\tau_k$. The complete variational action also carries the Berry form and both counterterms.

These are actual clock-threshold records, separately typed from original executed Writes and the eight partition addresses. Their source-to-Write identification is the physical instrument still to specify. The same construction applies to all six achieved clock certificates.

9. A pulse whose boundary returns

The retained regular-interior/outgoing-exterior wave action supplies a concrete account of transport. Choose one normalized monopole, positive R, κ, c, and a prescribed clamped source pulse $J = q$ on $[0, \tau_p)$, zero otherwise. Let $H = 2R/c$. [6]

Its exact impulse kernel and boundary coordinate are

$$g_0(t) = \frac{c}{R} 1_{(0,H)}(t),$$
$$a(t) = \frac{qc}{\kappa R} \operatorname{length}([0,\tau_p] \cap [t-H, t]).$$

The overlap rises, may plateau, and returns to zero at $\tau_p + H$. The bare ON coupling work is zero because $a(0) = 0$. Release supplies the exact complete cycle work

$$W = \frac{q^2}{\kappa R} \min\left(\frac{c\tau_p}{R}, 2\right) > 0$$

for nonzero q. An inherited stored counterterm contributes at ON and release and telescopes over the full OFF-to-OFF cycle. Both endpoint payments remain in the account.

This is a prescribed-control mean-energy example under one common stationary Gaussian reference. It retains its control cost and does not replace the autonomous microscopic source evolution.

10. Where the retained pulse resides

The boundary OFF energy is

$$E_b(t) = \frac{\kappa a(t)^2}{4R}.$$

The full field energy while ON is $qa(t)/R$; after release it is W. At every time,

$$E_{\mathrm{field}} = E_b + E_{\mathrm{radial+kinetic}},$$
$$E_{\mathrm{radial+kinetic}} \ge 0.$$

After $\tau_p + H$ the boundary mean is zero. Deterministic forcing preserves the stationary Gaussian covariance, so its whole reduced boundary marginal then matches the unforced reference. The outgoing bulk mean field still carries W. Independent spatial integration of that pulse gives the same energy.

For the declared normalization $R = \kappa = c = q = 1$ and $\tau_p = 1$, the boundary coordinate rises to one, falls to zero at $t = 3$, and $W = 1$. Its peak boundary energy is $1/4$. At late times all unit energy is in the bulk complement. These numbers illustrate the exact formula; they assign no physical source scale.

Persistent source memory is a second form of retention. In the inherited source law the written f_H contribution remains in J and continues to sustain a displaced field. A finite response horizon therefore need not erase either an outgoing account or a persistent source record. [6]

11. The spin complement and memory

The latest spin calculation completes J21, F11 and the retained T5 for all 54 preparations. It supplies a quantitative complement rather than an unspecified missing account. [7]

Let $v = Aw_d$ and let Π project onto a specified retained source span. The source inner product gives

$$v = \Pi v + r, \qquad \langle \Pi v, r\rangle = 0,$$
$$\|v\|^2 = \|\Pi v\|^2 + \|r\|^2.$$

The completed source-action excess uses the six-function span $(d, e, u_d, u_e, w_d, w_e)$. The next lower-response direction instead uses the five-function span ending at w_d. Their residuals retain their separate projections. Physical energy conversion retains the actual energy operator B:

$$\langle v, Bv\rangle = \langle \Pi v, B\Pi v\rangle + \langle r, Br\rangle + 2\,\mathrm{Re}\langle \Pi v, Br\rangle.$$

The mixed term remains unless the receiving projection is B-orthogonal.

A finite source also gives an exact dynamical continuation. Write $u' = -Au$, $p = \Pi u$, $q = (I - \Pi)u$. Solving the q equation and substituting it into p gives a retained evolution with a memory integral and an initial-hidden-state term. Eliminating a coordinate changes the evolution law unless those contributions vanish. The derivation is retained in the technical packet.

12. The cost of a specified record

Select one common positive source energy B and a linear receiver $r = L^T y$. The positive displacement cost is $y^T B y$. Put $G = L^T B^{-1} L$. For an attainable record,

$$\min_{L^T y = r} y^T B y = r^T G^{+} r.$$

The minimizing vector is $B^{-1} L G^{+} r$. Every other feasible vector adds a nonnegative B-norm. On constrained microscopic source states this ambient minimum remains a lower bound; equality requires that vector to be physically realizable.

The inherited literal eighteen-coordinate energy and a declared diagnostic receiver give an exact executable instance. This receiver is separately labelled from the original complete receiver family. [8]

A positive displacement cost and a signed energy change have different references. Relative to y_0,

$$\Delta E = 2 y_0^T B\, \delta y + \delta y^T B\, \delta y.$$

The native example has positive displacement cost $\frac{2006049}{573049}$ and signed energy change $-\frac{1017407}{573049}$. The mixed reference payment explains the signs.

For an ensemble, the complete mean energy additionally retains $\mathrm{Tr}(B\,\mathrm{covariance})$. A source-dependent projection minimum, a common reference energy, and the universal floor therefore keep their distinct types.

13. The structural finite ledger

The structural account receives distinction arity $h = 2$ and spatial support $d = 3$. Their fixed-binary closure $d^{d-1} = 2^d + 1$ has the unique integer solution $d = 3$ for d at least two. At $d = 2$ it fails; for d at least four, d^{d-1} is greater than $2^d + 1$.

The radix and bounded product kernel are

$$\mathscr{R} = h^2 d = 12,$$
$$\mathscr{B} = \{h^a d^b : 0 \le a \le 3,\ 0 \le b \le 2\}.$$

Its full grid is

a	$b=0$	$b=1$	$b=2$
0	1	3	9
1	2	6	18
2	4	12	36
3	8	24	72

Retaining values at most twelve gives $P = \{1,2,3,4,6,8,9,12\}$. The smallest admitted overflow is $\Theta = 18$. Sixteen requires exponent $a = 4$ and lies outside this kernel. The source caps and route grammar are part of the construction. Ordinary multiplication alone is not a complete physical Write law. [9]

The completed-WRITE route also supplies $\mathscr{R} = M_w D_{\mathrm{route}} = 4 \times 3$. Route depth and spatial support have distinct meanings, and its boundary assumptions remain attached. [9]

14. Carrier shares and the floor

The square account is $N_c = \mathscr{R}^2 = 144$. Its carrier and retained complement are

$$\Theta = 18 = \frac{N_c}{8}, \qquad N_{\mathrm{ret}} = 126 = \frac{7 N_c}{8}.$$

The binary face-state route assigns one unresolved carrier state among 2^d states. This explains its structural one-eighth share. A source energy fraction additionally requires the energy and receiving operators developed above. The closed ledger gives $162 - 18 = 144 = 126 + 18$. [9]

The floor combines separately typed compact cycle, arity and support:

$$A_0 = \frac{1}{(2\pi) h d} = \frac{1}{\pi \mathscr{R}} = \frac{1}{12\pi}.$$

The equality between $2h$ and h^2 uses $h = 2$. The source also registers the spherical projection $A_H/(4\pi d)$, with unit local closure $A_H = 1$. Its angular allocation and geometry remain explicit. [9]

One conditional energy receiver is

$$A_{\mathrm{rec}} = A_0 + \frac{E_{\mathrm{disp}}}{E_*}, \qquad E_* > 0.$$

Here E_{disp} is a selected positive common-reference displacement cost. This equation supplies a candidate physical interpretation of lift once its source incidence and scale are selected. It supplies neither a radial law nor a new numerical determination of A_0. Preparation-dependent spin norms remain separately normalized.

15. The common source and its two-level lanes

Keep the signed-response K_r in $Q = T + g_H K_r$ distinct from the commuting pair-energy operator K_e. Let $\alpha = \varepsilon_{\mathrm{phys}}/\hbar$ be the positive physical frequency scale. One complete unequal-occupation lane has the same Hamiltonian, full field, counterterms and evolution as a two-level source with those same operands. The common source supplies its further relations across lanes and occupation sectors. [2,11]

Removing only $Q_L Q_R$ from the complete pair Hamiltonian gives the additive control $S + Q_L + Q_R$. The removed term is scalar within each driven lane, so all six inherited current clocks are identical under this control. A complementary receiver reads its intersector energy instead.

The protected diagonal states have $D = C = 0$, including their source counterterms. Their exact frequencies are

$$\omega_{ba} = \alpha(q_b - q_a)(q_b + q_a + 2).$$

The same receiver in the additive control gives $2\alpha(q_b - q_a)$. The proposed three-state coherent preparation produces two adjacent frequency terms, with the same amplitude and one common scale. The source preparation and its interference receiver remain explicit physical inputs.

16. A uniform comparison and shared calibration

Order the four roots $q_0 < q_1 < q_2 < q_3$. In placement order $(1,0,2,3)$, $Q(g_H)$ is an irreducible real Jacobi matrix for every g_H at least zero. Its roots are simple; diagonal dominance gives

$$68 + 544 g_H \le q_a \le 100 + 2112 g_H.$$

Let $R_{\mathrm{pair}} = \omega_{21}/\omega_{10}$ and let R_{add} be the corresponding additive ratio. Then

$$\frac{R_{\mathrm{pair}}}{R_{\mathrm{add}}} = \frac{q_2 + q_1 + 2}{q_1 + q_0 + 2} > 1 + \frac{1}{132}.$$

The bound follows from interlacing, $q_2 - q_0$ at least $8 + 32 g_H$. It compares the two sources at the same g_H. A specified compact calibration interval gives a sharper margin. It does not assign every parameter the same sampling cadence or distinct adjacent frequencies.

The six bare signed couplings supply

$$\begin{aligned} \Xi &= \sum_{a<b} \frac{g_{ab}^2}{\Delta^2} \\ &= 384 + 5120 g_H + 1377280 g_H^2. \end{aligned}$$

This function is strictly increasing for g_H at least zero. Within the retained source law it identifies g_H from a dimensionless readout; one absolute scale then calibrates ε. The protected coherence ratio can be retained as a separate prediction rather than independently adjusted.

17. Receiving the hidden source offset

Current-lane couplings determine root differences. A complementary coherence ratio can also recover their common offset. Put

$$a = q_1 - q_0, \qquad b = q_2 - q_1,$$
$$\rho = R_{\mathrm{pair}}/(b/a).$$

Then

$$q_0 = \frac{1}{2}\left[\frac{a+b}{\rho - 1} - a - 2\right].$$

Three oriented couplings along a spanning tree, their common bare gap, and this ratio consequently reconstruct the four source eigenvalues in the declared model. Placement eigenvectors remain a separate receiving account. This inverse is an algebraic identity; its denominator shows the increasing sensitivity as ρ approaches one. [11]

Source access also has structure. Either endpoint placement in the Jacobi path reaches all four modes for every nonnegative g_H. Interior placement 0 instead misses one mode exactly at

$$g_H = \frac{1}{164}.$$

Native exact closure returns dimension three there, while endpoint placement 1 returns four. Thus source observability depends on preparation even though the Hamiltonian remains nondegenerate. This new exception was followed from Duck's first-round cyclicity breadcrumb.

18. Eight expectation values and a rank test

An autonomous closed two-state source, fixed preparation and fixed linear receiver have equally spaced expectations

$$y_n = c + a\cos(n\vartheta) + b\sin(n\vartheta).$$

Form first differences $d_n = y_{n+1} - y_n$ from eight means and the four-by-four difference-Hankel $H_{ij} = d_{i+j}$. The constant disappears and the two exponentials give rank at most two. A nonzero three-by-three minor distinguishes the protected source record from this class. Eight means are not eight individual shots. [11,12]

The earlier cadence supplied a determinant separated from zero at a per-mean allowance of 0.0001. Source-guided cadence selection now allows 0.01 under the same independently declared calibration family. It trades a longer coherent evolution for fewer repeated preparations; the physical stability cost stays explicit.

The next sections state the exact retained-memory recipient before specifying the preparation, readout, errors and shot counts for this test. Driven or open two-state sources and arbitrary multilevel sources remain distinct comparison classes.

19. Eliminate the hidden block exactly

Let A be the original finite self-adjoint positive generator, Π the retained orthogonal projection, and $Q = I - \Pi$. On retained and hidden coordinates p, q, write $B = \Pi A \Pi$, $C = QA\Pi$, $D = QAQ$. Then

$$\dot{p} = -Bp - C^* q, \qquad \dot{q} = -Cp - Dq.$$

Variation of constants gives

$$q(t) = e^{-Dt} q_0 - \int_0^t e^{-D(t-s)} C p(s)\, ds.$$

Substitution retains both the initially hidden state and its feedback:

$$\begin{aligned} \dot{p} &= -Bp - C^* e^{-Dt} q_0 \\ &\quad + \int_0^t M(t-s) p(s)\, ds, \\ M(t) &= C^* e^{-Dt} C. \end{aligned}$$

For initially retained data, differentiating twice gives $\Pi A^2 \Pi = B^2 + C^\dagger C$. Thus equality of compressed evolution and $\exp(-Bt)$ for every retained preparation forces $C^\dagger C = 0$, hence $C = 0$. Conversely $C = 0$ decouples the equations. Exact autonomous closure is therefore equivalent to invariance $AV \subseteq V$. [12]

For v in V, its initial memory strength is exactly $\|QAv\|^2$. Memory comes from the actual omitted source directions, not from a fitted delay term.

20. Normalize the residual in its source Gram

Let b be a nonorthonormal retained basis with Gram $G > 0$. Put D_b and T_b equal to the source inner-product matrices shown

below. Projection gives

$$D_b = \langle b, Ab\rangle, \quad T_b = \langle Ab, Ab\rangle.$$
$$L_b = T_b - D_b G^{-1} D_b,$$
$$K_0 = G^{-1/2} L_b G^{-1/2} = C^* C$$

in an orthonormal chart. $\mathrm{Rank}(L_b) = \mathrm{rank}(C)$; the physical memory norm is the largest generalized eigenvalue of $L_b v = \lambda G v$. Ordinary eigenvalues of the unnormalized residual matrix do not supply that norm.

Suppose the original invariant exchange-odd sector has $\gamma I \le A \le NI$. Apply variation of constants once more to the retained equation with $q_0 = 0$. The evolution defect is a double integral with factors $\exp[-B(t-s)]$, $C^\dagger$, $\exp[-D(s-r)]$, C and $\Pi\exp(-Ar)$. Their three time intervals sum to t. Consequently

$$\left\| \Pi e^{-At}|_V - e^{-Bt} \right\| \le \frac{t^2}{2} e^{-\gamma t} \|K_0\|.$$

Using $A \le NI$ also bounds the difference from its quadratic term by $Nt^3\|K_0\|/2$. These are original A-time bounds; the expected attempt count is Nt. A physical time calibration remains separate. [12]

21. Continue the actual spin closure

The completed five-function spin span equals

$$V_5 = \mathrm{span}(d, e, Ad, Ae, A^2 d).$$

Leading-one residual coordinates make the first enlargement exactly

$$V_7 = \mathrm{span}(V_5, A^2 e, A^3 d).$$

The earlier all-54 preparation certificate established $\dim V_7 = 7$. The next two possible outgoing functions are $A^3 e$ and $A^4 d$. A literal full-source point certificate computes their rank rather than assuming the enlargement closes. [7,11,12]

All 54 evaluated original finite-N preparations have a paired Krylov rank of at least 10. The seven-dimensional enlargement has exact residual rank two in every evaluated preparation. Selected deeper certificates reach rank 14. The retained rank has not stabilized: these are exact full-source closure lower bounds, with strict positive Gibbs residual bounds and an explicit finite invariant count-space upper closure. No exact minimum or uncomputed global residual moment is assigned.

All original sites, Gibbs weights, self-excluded fields, accepted flip rates and attempt clocks remain present. The evaluator lumps only exactly exchangeable background-site classes and retains their exact multiplicities. This is a full finite-source calculation, distinct from the earlier conditional twenty-site contraction.

The minimum invariant closure is the Krylov union $V + AV + A^2V + \cdots$. Exact rank stabilization, or a source minimal-polynomial identity, is the stopping condition.

In the deeper selected source, the natural retained dimensions advance $5, 7, 9, 11, 13$, each with a nonzero outgoing residual. The fourteen-column certificate makes V_{13} nonclosed because $A^6 e$ remains independent. Also $M_7(t) = C_7^\dagger \exp(-D_7 t) C_7$ has rank two at every finite time: the hidden exponential is strictly positive in this finite source. This is a kernel-rank statement, not a rank statement for the propagation error.

22. Compact certificates and an invariant upper span

Evaluate m source functions at m literal configurations and form M. A nonzero certified determinant establishes their independence pointwise. Every configuration has positive mass in the original finite Gibbs law, so this is independence in that same physical source Hilbert space.

For $M = [V_7, A^3 e]$, the last row λ of M^{-1} annihilates V_7 and reads $A^3 e$ as one. For $r = Q_7 A^3 e$,

$$1 \le \left[\sum_i \frac{|\lambda_i|^2}{\pi(x_i)} \right] \|r\|^2.$$

If the original Gibbs log weights lie within plus or minus L after a common constant is removed, $\pi(x) \ge \exp[-N\log 2 - 2L]$. Hence

$$\|r\|^2 \ge \frac{e^{-N\log 2 - 2L}}{\sum_i |\lambda_i|^2} > 0.$$

Two inverse witness rows similarly give a positive two-by-two residual-Gram lower bound. This certifies nonclosure in the original Gibbs norm, without assigning a useful approximation size to that very conservative lower bound. [12]

An explicit invariant upper space retains all labelled twenty-site core configurations and counts each exchangeable background block:

$$D_{\mathrm{lump}} = 2^{20}(F_b + 1)\binom{P_b + 2}{2}\binom{T_b + 3}{3}.$$

Its dimensions have 96 to 342 decimal digits across the 54 original preparations. This finite upper closure does not assert that the d/e seeds reach it.

23. The nonzero-clock reference account

The inherited amplitude bound constrains a simple occupation-compensating reference. Put $r = 1 - n$. If actual clock amplitude is at least $m > 0$,

$$\frac{1 - \sqrt{1 - m^2}}{2} \le r \le \frac{1 + \sqrt{1 - m^2}}{2}.$$

The explicit reference state

$$|r\rangle = \sqrt{n}\,|0\rangle + \sqrt{1 - n}\, e^{-i\gamma}|1\rangle$$

has connection $(1 - n)\, d\gamma$, even when n changes differentiably: its real amplitude-derivative terms cancel. It restores the occupation coefficient in the receiving lift while keeping the joint-state Berry form. It does not automatically receive that remaining form. [11]

For actual lane 01 this reference occupation lies approximately between 0.499975039 and 0.500024961. This is a recipient-specific action bound. The next construction instead receives the entire actual joint-state contribution with a balanced reference and a measurable relative phase.

Finite-energy overlap derivatives suffice for the actual continuum trajectory. No additional assumption that its initial vector belongs to the strong operator domain is inserted.

24. A reference retaining the actual energy

Retain the unchanged Chapter I joint H and its preparation Ψ_0. Its conserved form energy E includes the even source energy, full field, both counterterms and joint correlations. For prescribed legal phase rates, put

$$w = \dot\gamma + 3\dot\chi, \qquad h_R = 2(w-E)P_R.$$

The balanced reference has relative phase $\dot\beta = 2(w-E)$. The product trajectory retains precisely the original source-field marginal. Its canonical connection is

$$E\,d\tau + \frac{1}{2}d\beta = d\gamma + 3d\chi.$$

For the actual lane 01 preparation, $H = \varepsilon e_{01} I + \varepsilon S + K$, with $K \geq 0$ and preparation expectation of K at most K_{prep}. Therefore

$$\varepsilon e_{01} - \varepsilon \leq E \leq \varepsilon e_{01} + \varepsilon + K_{\mathrm{prep}}.$$

The exact rational certificates enclose E within $[0.917629340281369063, 0.917632830384309734]$. Equivalently $E = \varepsilon e_{01} + (E_0+E_1)/2 + c_{\mathrm{soft}}$ for the retained hard branches and full soft vacuum. This uses the actual continuum preparation, rather than the earlier finite-qubit diagnostic. Physical energy is $\hbar E/t_u$. [2,12]

25. One original forward/reverse/wait protocol

Keep activation $B = 1$, $A_{\mathrm{path}} = 0$ and the declared diagnostic source ratio $R_H/(ct_u) = 1/10$. R_H is the original source radius; its incidence with the field radius stays explicit. Starting at 000, apply W_1^+ to 100, W_1^- back to 000, then wait four model-time units.

The lifted $\phi_b = \phi_p = 0$. Thus $\gamma = \delta = \phi_a$ rises from 0 to $\pi/2$ and returns to 0; $\chi = 0$. Original signed actions are $+\pi\hbar/2$, $-\pi\hbar/2$ and zero. With ℓ increasing along either traversal,

$$f(\ell) = \frac{12\sqrt{68 - 32\cos\ell}}{100 - 32\cos\ell + 4\cos^2\ell},$$
$$d\tau = \frac{f(\ell)}{10}d\ell, \qquad v = 10/f.$$

Both active durations equal $T = 0.158555895876030285\ldots$. The forward integral is retained from H02; H05 supplies its reverse after a common-phase shift. Native intervals overlap, and exact path reversal identifies their positive durations. [12]

The general source-paced instrument uses $w = \sigma v_j$, including its original positive path clock. A phase-threshold receiving schedule agrees with it only when $v_j = (M/3)\nu(t)$. Signed role phase, elapsed source time and actual current phase remain separate operands.

26. Entry, matching and release stay distinct

Start the reference balanced with $h_R = 0$. Over external entry $a = 0.01$, hold 000 and ramp $h_R = 2(10-E)(s/a)P$. During the forward/reverse/wait intervals use $2(w-E)P$. Apply the recorded sudden changes at event boundaries. After the wait, release over $b = 0.01$ with $h_R = -2E(1-s/b)P$. These ramps are outside ledger matching.

Event	τ	β
Start	a	$.1 - Ea$
Forward	$a+T$	$.1 + \pi - E(a+2T)$
Reverse	$a+2T$	$.1 - E(a+4T)$
Wait	$a+2T+4$	$.1 - E(a+4T+8)$
Release	$a+2T+4+b$	$.1 - E(a+b+4T+8)$

Source wait has $w = 0$ and $h_R = -2EP$; it is not reference off. The wait winds β by $-8E$, more than one full turn. Actual θ advances throughout and is enclosed at every boundary by $[\ell_{01}\tau_{\mathrm{lower}}, u_{01}\tau_{\mathrm{upper}}]$.

The final τ is $4.337111791752060571\ldots$. Final actual θ lies in $[0.000015136966570142, 0.000015136966591669]$, while β lies in $[-7.841399681709530363, -7.841369477578352354]$. The two phases record different quantities. [12]

27. Every switch and continuous-drive payment

On the balanced reference, energy is $w-E$ and variance is $(w-E)^2$; mean drive work is dw. The original path has $v_0 = 10$, $v_L = 125/(3\sqrt{17})$, and $v_{\min} = 9.810320323234103\ldots$ at $\cos\ell_* = (11-\sqrt{73})/4$. Its interior minimum splits each drive into two monotone pieces. [12]

Reference work piece	Work in $\hbar/t_u$
External entry	$10 - E$
Forward first piece	$v_{\min} - 10$
Forward second piece	$v_L - v_{\min}$
Forward to reverse	$-2v_L$
Reverse first piece	$v_L - v_{\min}$
Reverse second piece	$v_{\min} - 10$
Reverse to wait	10
Wait	0 drive work
External release	E

The signed sum is zero, but positive and withdrawn work each equal $2v_L + 2(10 - v_{\min})$, approximately 20.590661439892877. Recoverable exchange and irreversible apparatus cost are distinct accounts.

Adding a nonnegative-Hamiltonian baseline adds canonical payment $C = \pi + 2ET + 8E + Eb$. The total variational form

still subtracts energy and vanishes on its own exact driven trajectory. Switching cancellation supplies no zero-cost apparatus claim.

28. Read the phase and keep its turn counter

Independent prefix preparations measure $X = \cos\beta$ and $Y = -\sin\beta$. Use atan2$(-\widehat{Y},\widehat{X})$, then nearest-branch continuation from $\beta_0 = 0$. An endpoint qubit alone records phase modulo 2π; an external counter keeps its turns.

Include all boundaries and a grid with gaps at most 0.01: no more than 441 prefixes. Since $E < 1$ and $v \le 50/3$, $|\dot\beta| \le 106/3$. Coordinate errors $\le 1/16$ imply angular errors $< 1/8$ rad. Consecutive true advances plus both angular errors obey

$$53/150 + 1/4 < \pi,$$

so the unwrapped branch is unique. Correct branch choices leave each prefix error below $1/8$ without summing all prefix errors into drift.

Allocate coordinate tolerance as systematic $1/32$ plus statistical $1/32$. Four systematic budgets of $1/128$ cover reference preparation, evolution/waveform, readout and timing. Sufficient conditions are trace-distance and Duhamel bounds $\le 1/256$, Pauli-readout norm error $\le 1/128$ and trigger error $\le 3/13568$ model time.

With 26,624 independent shots per coordinate and prefix, the simultaneous statistical failure is below 1%. Maximum fresh preparations are 23,482,368. Each readout disturbs that copy; fresh prefixes retain the history. Source energy and continuum preparation require their own retained energy-form control. [12]

29. Specify the protected-coherence instrument

Write $|a\rangle$ for the protected state $|q_a q_a\rangle$. Prepare

$$|\psi\rangle = (|0\rangle + |1\rangle + |2\rangle)/\sqrt{3}$$

with the field vacuum. D,C and their counterterms annihilate this protected span under the full retained Hamiltonian. The adjacent-coherence receiver is

$$O = |0\rangle\langle 1| + |1\rangle\langle 0| + |1\rangle\langle 2| + |2\rangle\langle 1|.$$

Its outcomes are $+\sqrt{2}, -\sqrt{2}, 0$. Corresponding vectors in the three-state basis are $(1,\sqrt{2},1)/2$, $(1,-\sqrt{2},1)/2$ and $(1,0,-1)/\sqrt{2}$; complementary outcomes report zero. A source-basis rotation followed by this projective readout specifies a mathematical instrument. [12]

Its expectation is $y(t) = (2/3)[\cos(\omega_{10}t) + \cos(\omega_{21}t)]$. Every sampling time receives fresh preparations. Measuring one copy changes its state and cannot supply its next undisturbed expectation.

The required physical coherent mode access, receiver pulse, field preparation, reset and cooling belong to the apparatus. The complete error budget below assigns their permitted effects before using a sample rank.

30. Charge leakage, disturbance and timing

Use trace distance $D(\rho,\sigma) = \|\rho - \sigma\|_1/2$. Preparation distance d_{prep} gives expectation bias $\le 2\sqrt{2}\, d_{\mathrm{prep}}$. Leakage l and conditional error d_{cond} satisfy

$$d_{\mathrm{prep}} \le \sqrt{l} + d_{\mathrm{cond}}.$$

Leakage alone does not specify the state inside the protected space or its field preparation. An effective measured observable M with $\|M - O\| \le b_{\mathrm{read}}$ includes pulse disturbance and outcome assignment in its readout bias.

Unwanted dynamics contributes at most $2\sqrt{2}\,\kappa$, where

$$\kappa = \int \|\delta H(s)\psi_{\mathrm{ideal}}(s)\|\, ds/\hbar.$$

This bound extends through the latest allowed trigger. A source-only finite-matrix norm does not bound an unbounded field interaction.

The retained independent calibration is $\Xi = 1382784$ plus or minus one and $\alpha/\alpha_{\mathrm{nominal}} = 1$ plus or minus one part per million. It gives $|y'| \le L$ with

$$L < 114081.396605\,\alpha_{\mathrm{nominal}}.$$

Trigger error δ_t contributes $L\delta_t$. Calibration uncertainty is already part of the ideal source family; it is not charged again as timing jitter. [12]

31. A cadence selected from the source

Before acquiring the target record, choose

$$h = \frac{5\pi}{2\alpha_{\mathrm{nominal}} D_{\mathrm{nominal}}},$$

$$D_{\mathrm{nominal}} = \frac{6348467821812271}{80000000000},$$

and $t_k = kh$ for $k = 0,\dots,7$. This h is 7.5 times the earlier cadence and was selected from a declared finite list using only the retained source. Both sampled frequencies remain distinct and nonaliased. For the three-by-three difference minor A with index set $I = \{0,2,3\}$,

$$5.318723982743267 < \det A < 5.941782789331066,$$

$$|A_{ij}| < 2.$$

For per-expectation allowance $\eta = 0.01$, the conservative two-record determinant perturbation bound is

$$72\eta(2+4\eta)^2 = 2.996352,$$

below the certified lower determinant. This source-selected cadence exchanges a longer stability interval for substantially fewer shots. It is not fitted to measured target data. [12]

Allocate 0.001 each to preparation, unwanted evolution, effective readout and timing, leaving 0.006 for statistical error. The next section turns this allocation into explicit measurement counts and a decision rule.

32. Repetitions and an operational decision

Sufficient apparatus conditions are $d_{\mathrm{prep}}, \kappa \le 0.001/(2\sqrt{2})$, $\|M-O\| \le 0.001$ and $\delta_t \le 8.76567 \times 10^{-9}/\alpha_{\mathrm{nominal}}$. If conditional preparation is exact, coherent leakage $l \le 1.25 \times 10^{-7}$ suffices. These are requirements rather than achieved precision. [12]

For N independent outcomes in $[-\sqrt{2}, \sqrt{2}]$ and eight means, Hoeffding and a union bound give

$$P(\text{any mean error} > s) \le 16e^{-Ns^2/4}.$$

At $s = 0.006$ and failure allowance 0.01, $N = 819{,}751$ fresh shots per time suffice: 6,558,008 total. Calibration failures add their probability to that sampling allowance. The earlier 0.0001 allocation would need 8,197,509,899 shots per time under the same split.

Form measured differences, $\widehat{A}_{ij} = \widehat{d}_{(I_i+I_j)}$, and $\widehat{B} = \max|\widehat{A}_{ij}|$. Reject the stated closed two-state class when

$$|\det \widehat{A}| > 36\eta(\widehat{B}+2\eta)^2.$$

Each entry error is $\le 2\eta$. Telescoping each product of three entries across six determinant terms gives this threshold because every ideal two-state minor vanishes. The calibrated target exceeds it by more than 2.35160 on the stated simultaneous sampling event, conditional on the apparatus bounds.

33. The minimum counter count

For polynomial phase receivers through degree d, retain $M_{jk} = \int \theta^k 1_j \, d\theta$, and the corresponding event-atomic sums when required. A segment α to β adds $(\beta^{k+1} - \alpha^{k+1})/(k+1)$, clipped to each specified gate. Actual phase endpoints suffice; no constant-rate approximation is used. [11]

Here is the rank derivation of minimality. Collect required additive receiver values into $F(h)$ and subtract $F(h_0)$ when initial endpoints and total duration/count are supplied. Let V be the span of these contrasts with dimension r. A summary with c linear counters and a linear decoder must be injective on V; otherwise a nonzero receiver contrast disappears. Rank-nullity therefore gives $c \ge r$. Coordinates in any basis of V attain r.

Known endpoint identities reduce r. For fixed event addresses $\theta_n = hn$ and N events, summing labels at each degree gives $h^k \sum n^k$, already determined by N. Overlapping gates or permitted source histories can supply further constraints.

This is exact minimality for the admitted linear additive receiver class. Online continuation also retains predictive state and current phase; a finite-family contrast basis alone is not a transition-closed realization for every future history.

Counter dimension does not bound precision or stored bit count, and makes no minimum claim for unrestricted nonlinear memory.

34. ABBA/BAAB with actual event offsets

Use the original closed blocks $A = (W_1^+)^4$ and $B = (W_7^+)^4$. Every block returns to 000. In one repeated source context their positive durations are T_A, T_B; retain each internal event offset u. The A-block starts in ABBA are $\{0, T_A + 2T_B\}$; in BAAB they are $\{T_B, T_B + T_A\}$. Their sums agree, while

$$\sum_{ABBA} (s+u)^2 - \sum_{BAAB} (s+u)^2 = 2T_B(T_A + T_B).$$

Expansion cancels both $2u\sum s$ and $2u^2$. B-block starts similarly have equal sums, with squared-sum difference $-2T_A(T_A+T_B)$. Each of three A-only output points occurs once per A; the common point occurs once per A and four times per B. Its difference is consequently $2(T_A+T_B)(T_B-4T_A)$. [11]

Counts, endpoints and first physical-time atomic moments agree, while second moments distinguish the histories even with unequal durations. Under the prescribed equal phase-step schedule the A-only differences are $64h^2$ and the common-point difference is $-192h^2$.

Prouhet closed-block reorderings extend the phase-event distinction beyond every finite polynomial degree. Nonlinear transport from physical time to actual phase retains original time arguments; it does not automatically preserve unweighted phase moments.

35. An attainable source record has a price

For positive B and record $r = L^\dagger y$, let $y_\star$ be its ambient energy minimizer and $G = L^\dagger B^{-1} L$. If F_r is the permitted microscopic displacement set, projection orthogonality gives

$$\min_{y \in F_r} y^* B y = r^* G^+ r + \min_{y \in F_r} \|y - y_\star\|_B^2.$$

Indeed $L^\dagger(y - y_\star) = 0$, and $By_\star$ lies in range(L), so the cross term vanishes. The second term is the realizability correction. A finite nonempty permitted set attains its minimum; an empty compatible set has no allowed endpoint. [11]

For the original eighteen-coordinate literal energy and fixed outside spin configuration, permitted port displacements are $(0, 0, e_z - e_0)$. Every signed cross term is retained. The diagnostic record is $(-1, 0)$ under the two declared receiver columns, distinct from the complete original receiver family.

Payment / κ_E	Value
Ambient minimum	0.987626515
Pure-source minimum	1.894317938
Realizability correction	0.906691423

The source minimum is $1085537/573049$. The complete endpoint enumeration supplies its certificate in the next section.

36. Every endpoint and the feasible minimum

The diagnostic reads the e_0 and e_1 coordinates. A displacement from z_0 has record $(-1,0)$ exactly for $z = 2,\dots,15$. The table gives numerator of cost/κ_E over denominator 573049; z_{15} has numerator 1146098. [11]

z	Record	Numerator	One flip?
0	$(0,0)$	0	reference
1	$(-1,1)$	2006049	yes
2	$(-1,0)$	1085537	yes
3	$(-1,0)$	1146097	no
4	$(-1,0)$	2006049	yes
5	$(-1,0)$	4965618	no
6	$(-1,0)$	2155185	no
7	$(-1,0)$	2163505	no
8	$(-1,0)$	1085537	yes
9	$(-1,0)$	2155185	no
10	$(-1,0)$	1142898	no
11	$(-1,0)$	1146097	no
12	$(-1,0)$	1146097	no
13	$(-1,0)$	2163505	no
14	$(-1,0)$	1146097	no
15	$(-1,0)$	1146098	no

Thus z_2 and z_8 attain the smallest cost on the nonzero record fiber $(-1,0)$; both are single accepted port flips. z_0 and z_1 cannot realize this record. Outside-coordinate changes add nonnegative squares and cannot lower the attained fixed-outside minimum.

Mixtures retaining this exact record obey the same endpoint floor for expected quadratic payment. Energy of mean displacement is a different quantity and retains covariance. Preparation and physical κ_E remain separate apparatus assignments.

37. Calibrate the paid protocol

The ideal paid reference uses actual E. The present source certificates also give a numerical control: take their exact rational energy midpoint $\widehat{E}$ and replace E by $\widehat{E}$ in every reference waveform. Then

$$|E-\widehat{E}| < 1.746\times 10^{-6}.$$

The total connection on matching intervals becomes $[w+(E-\widehat{E})]\,d\tau$. Across the complete forward/reverse/wait program its signed-ledger bias is below $7.54\times 10^{-6}\hbar$. The source energy stays E; its calibration residual remains visible. [12]

The entry and release canonical integrals are $(E+10)a/2$ and $Eb/2$ for exact calibration. They are external to the signed Write ledger. A reconstructed matching action with interval U, energy error dE, time error dt and endpoint angular errors η obeys the conservative allowance

$$U\,dE + (|E|+dE)\,dt + \eta.$$

An actuator using a finite-time turnaround in place of a sudden quench adds its own interval, phase and work. It cannot silently reuse the ideal endpoint account. Actual clock certificates continue to apply only while its original source-field H and preparation are retained.

38. Pay for the controller

The specified waveform is an externally driven instrument. A material autonomous controller must produce its schedule, store or recover work and retain the phase-turn record. When coupled to the quantum source or reference it can also become entangled; the enlarged Hamiltonian then determines the clock and energy account.

The protocol has made the required work exchange finite and explicit. It has not assigned a cost to original source actuation, waveform generation, qubit preparation, readout rotations, detector reset, external-counter memory or cooling. Their physical models supply those further payments. Recovering equal positive and negative reference work does not determine their dissipation.

For the protected-coherence experiment, the independent common calibration fixes g_H through Ξ and fixes ε through one absolute scale. The two-frequency record is then a prediction. Its source-guided cadence and apparatus tolerances specify what a proposed implementation must deliver.

These are concrete construction tasks with quantitative targets. They can receive hardware and channel assignments while preserving the original signed source response, phase lift, full field and energetic complement. [12]

The actual clock has a separate receiver. Divide its D and C expectations by signed $d_{01} < 0$ to obtain the complex current z; $\theta = -\arg z$. Its certified amplitude exceeds 0.999999998. Fresh source copies can estimate those two bounded Pauli coordinates, then unwrap $-\operatorname{atan2}(\operatorname{Im} z, \operatorname{Re} z)$. Their sampling and apparatus allowances are additional. The θ intervals in this protocol are predicted from retained actual-rate bounds; reference-qubit measurements read β.

39. The operation determines the retained account

A uniform A_0 remains in a total account while gradients and shared endpoint differences remove it. A raw lapse ratio instead rescales lifts through background normalization; a source-normalized local lapse declares a different input. Every readout retains its operation and reference. [1,9]

The source comparison sharpens this lesson. Current clocks leave the commuting pair-product energy unread. Protected coherence receives that energy, linked to the lane family by a common calibration.

Memory gives the dynamical version: a retained span evolves autonomously exactly when it is invariant. The actual seven-dimensional spin enlargement has further outgoing directions;

the new point certificates state how far its exact rank reaches. The invariant upper lumping retains the finite source without identifying it as the smallest closure.

History moments identify what a later receiver needs, and feasible endpoint costs identify an attainable payment. The paid reference now has one complete timed Write/inverse/wait program, all reference work pieces, a winding record and a fresh-copy measurement budget. Physical realization can build directly on these specified quantities.

40. Sources and reproducibility

The source package retains the original draft, completed Chapter I and prior Chapter II development sources. Editable typesetting, bundled fonts, exact operands, native receipts and independent offline checks accompany this reader. Every calculation retains its source preparation and units. [12]

[1] Owner Chapter II draft and verification ZIP, 15 October 2026.
[2] Final Chapter I and actual full-field clocks; H002628/H002630.
[3] Native history reduction; H002165.
[4] Observed-history clock/ruler recipient; H002559.
[5] Original Write/tensor/action map; H002623.
[6] Radiative memory and complete work; H002319.
[7] Completed spin source-action T5; H002627.
[8] Common spin energy and receivers; H002423.
[9] Original Chapter II packet CR001/002/003/005 and local CR113/114/115/218/238.
[10] Chapter II first development; H002634.
[11] Chapter II continuation, two 180-second Duck rounds; H002635.
[12] voli_chapter2_physical3_20261015: current source-bound physical-development packet. Source-bound physical-development packet from GEN6 Duck run chapter2-physical-protocols-20261015-01 (Duck 1.2.2, core 0.10.2; 15 native tasks), directed by Sean Brady; constructions written with OpenAI Codex (Courtroom II case voli_provenance/01, H002821).

41. Give the retained energy a spatial account

The completed endpoint minimum is a positive displacement payment for a selected record fiber. It is not the full energy of the source, its field or its controller. The interacting preparation has a separately retained conserved energy E, including its even source contribution and both counterterms. A gravitational mass assignment needs that complete reference account and a physical conversion. [13]

A spatial source requires more information still. Define an energy observable smeared over physical regions, identify its reference and units, and require its whole-space integral to recover the retained total Hamiltonian. Source, interaction, field and boundary payments need declared incidence. A positive global form does not automatically provide a positive pointwise density.

The classical commuting static-boundary chart gives a compact control. At its minimizing boundary value, put

$$A = \frac{\varepsilon^2 \sum_\alpha J_\alpha^2}{4R}.$$

The free field pays A, its interaction contributes $-2A$, and the counterterm contributes A. Their integrated sector sums to zero. Taking only the positive free-field density would omit two retained payments. This is not a joint static replacement of the noncommuting quantum D and C, and it does not remove the separate source energy. [13]

The original model chart converts displayed energy to $\hbar$ divided by the physical time unit. Neither that unit in seconds nor a physical ruler is fixed by the retained clock theorem. Writing E/c^2 before supplying these assignments does not derive kilograms or a spatial density.

The next source examples keep their energy, location and receiving operations typed separately. They calculate what follows from an explicit incidence rather than substituting a diagnostic record cost for the whole source.

42. What the recovered local effects retain

A retained Si29 effect construction has a useful exact consequence. Write $h = \sqrt{Q}$, $H_0 = \operatorname{diag}(h,h)$, and let O be its registered effects at channel t and site x. The energy effects

$$E_{t,x} = H_0^{1/2} O_{t,x} H_0^{1/2}$$

obey, for each of the eight sites,

$$\sum_t E_{t,x} = \frac{H_0}{8}.$$

Thus the channel-summed site energy is flat for every preparation, although the total expectation of H_0 can change with preparation. Resolving channels and changing the spatial profile are distinct operations. The effects alone do not identify physical site separation or a gravitating density. [13]

A specified receiving example places eight equal uniform balls at the vertices of a cube of side 2ℓ, with ball radius $\delta < \ell$. Assigning mass $M/8$ to each ball then makes the density explicit. M is one common physical energy conversion times the expectation of H_0, divided by c^2.

The cube embedding, smearing radius and energy-to-mass rule are declared assignments. They are not deductions from the flat site-effect identity, and the selected source account does not include otherwise unlocalized fields or controllers.

43. A spatial receiving operator with exact flux

A finite radial receiver makes the Poisson interface executable. Use radii ih, a zero lift at radius Lh, and edge conductances $i(i+1)$. With B the incidence matrix and C the diagonal conductance matrix, the positive Dirichlet operator is $K = B^{\mathsf{T}} C B$. Its exact inverse on $i, j = 1, \ldots, L-1$ is

$$H_{ij} = \frac{1}{\max(i,j)} - \frac{1}{L}.$$

For shell masses measured in one physical mass unit, put κ equal to twice G times that unit, divided by $c^2 h$. The source $q = \kappa m$ gives the lift $a = Hq$ and

$$i(i+1)(a_i - a_{i+1}) = \kappa \sum_{j \le i} m_j.$$

This is an exact mass-flux statement for the declared spherical shell recipient. It is not a claim that the cube vertices are those shells or that finite source-state closure derives spatial locality. [13]

Adding twice G times the total mass, divided by c^2Lh, converts the finite-boundary lift to the isolated infinity reference. It changes potential and energy reference while leaving interior gradients unchanged. For a uniform ball approximated by n shells, the reduced-lift error is below $3/n$; the sharper bound retains the harmonic sum. A finite outer boundary has its own separate error.

44. Geometry changes the same scalar reception

Outside the eight disjoint uniform balls, the cube density has exactly the potential of eight point masses at their centers. Equal weights and inversion symmetry cancel the dipole, quadrupole and cubic multipoles. The leading angular correction is of fourth order in ℓ/r. A spherical monopole is therefore a controlled far-field approximation, not an exact consequence of the total mass alone. [13]

The isolated binding account includes self and pair terms:

$$U_{\mathrm{self}} = -\frac{3GM^2}{40\delta},$$
$$U_{\mathrm{pair}} = -\frac{GM^2}{64\ell}\left(6+3\sqrt{2}+\frac{2}{\sqrt{3}}\right).$$

Shrinking the balls without declaring their self-energy treatment is not a finite-energy point-source limit. Both terms have a common reference and count each pair once.

Chapter I's quadrupole field also has energy outside its source boundary. Its exterior is not automatically a vacuum region for a gravitational density that includes that energy. The exact exterior $1/r$ law requires enclosing the selected density or controlling its remaining tail. A quadrupole amplitude, its positive energy, and a scalar gravitational monopole remain different objects.

45. Energy invisible to the actual current clock

Retain the original complete sixteen-state source, its commuting even energy, $S = \mathrm{Swap}$ and the unnormalized current $D + iC$. Exact source algebra gives

$$[S, Z_{\mathrm{raw}}] = -2Z_{\mathrm{raw}},$$
$$[E_{\mathrm{even}}, Z_{\mathrm{raw}}] = 0,$$
$$[D^2 + C^2, Z_{\mathrm{raw}}] = 0.$$

The even operator and complete counterterm also commute with the full original Hamiltonian and both field couplings. Adding a bounded function of either changes total energy but leaves every current history unchanged. A common scalar is the simplest example; it also changes the state's common phase. The independent original Write phase χ retains its own role. [13]

The four equal-occupation eigenstates have zero D and C but distinct energies $\varepsilon(1+q)^2$. They are controls for energy missed by the current receiver, not functioning current clocks. Source energy therefore cannot be inferred from current amplitude or phase rate alone.

On an active lane the actual receiver remains $z = \langle D + iC\rangle/(\text{signed } d)$, with $\theta = -\arg z$. Its phase requires separate source copies and its own preparation, timing, readout and winding budget. The reference qubit measures a different phase.

The separate retained Si29 transfer source gives another exact control. Its incoming-minus-outgoing population B follows by summing the original sector-resolved effects. Its positive excitation energy commutes with both B and the signed transfer generator. The proposed uniform excitation-density coupling therefore gives

$$H_a = H_{\mathrm{tr}} - \frac{a}{2}|H_{\mathrm{tr}}|, \qquad B_a(t) = B_0(t).$$

Its entire population-transfer history is unchanged. Full scaling of the signed generator instead dilates that history in time. This is the actual retained bright-sector signal, distinct from Chapter I's certified phase clock. Other species/site observables need not share its invariance. [13]

46. Couple a spatial lift to the complete clock

Write the full Hamiltonian as the sum of source, free-field, interaction and counterterm contributions. These retain the original complete source, both signed interactions and both counterterms. A supplied spatial lift must specify a family $H(a)$, rather than just a desired output frequency. [13]

For a fixed preparation and bounded perturbation V, Duhamel gives the first-order response

$$\delta z(t) = i\int_0^t \langle [V_I(s), Z_I(t)]\rangle\, ds.$$

The actual rate response is

$$\delta\nu = -\operatorname{Im}\left(\frac{\delta\dot{z}}{z} - \frac{\dot{z}\,\delta z}{z^2}\right).$$

An unbounded field form requires its energy-domain version and explicit propagation bounds. Changing the preparation supplies additional derivative terms. None is replaced by a finite matrix commutator alone.

The response target is shared across all six separately prepared original lanes, with one physical ε, ruler, time chart and perturbation. A stationary branch or matched internal age makes a fractional lapse test unambiguous. At equal coordinate time a nonstationary clock also carries its transient response. Independent fitting of each gap cannot test universality.

The fixed-field second-order diagnostic defines s as the logarithmic gap derivative with respect to ε. Its outward enclosures differ across all six lanes:

Lane	10^9 times $(1-s)$, lower	Upper
01	4.82	4.84
02	20.74	20.76
03	3214.92	3216.50
12	5.55	5.56
13	2970.53	2971.99
23	2719.14	2720.48

These are hard-branch perturbative diagnostics, with the original field fixed; they are not achieved physical redshift measurements.

47. A full scaling and its countercontrol

If the coupling is declared to be $H(a) = n(a)H_0 + f(a)I$ for static a, the exact full-field control is

$$z(t;a) = z(n(a)t;0),$$
$$\nu(t;a) = n(a)\nu(n(a)t;0).$$

It holds for all six lanes and includes every original field sector. It does not derive n from the source. At matched internal age the rate ratio is n. A coordinate-time derivative also contains t times the derivative of the original rate, where that extra regularity exists. Accumulated phase comparison needs only the retained continuously differentiable phase. [13]

Scaling ε alone while keeping the field fixed multiplies the source and interaction terms by n, the counterterm by n^2, and leaves the free field unchanged. At first order,

$$V_\varepsilon - n'(0)H_0 = n'(0)(H_{\mathrm{ct}} - H_f).$$

The counterterm part is invisible throughout the current history because it commutes with both H_0 and the receiver. The free-field part can change the clock through its interaction. The retained two-polarization hard-gap expansion gives six distinct ε-response diagnostics; these do not replace a full-field finite-lift error certificate.

The source-only square root and raw total-account square root also differ: background normalization retains the latter's sensitivity factor $1/(1-A_0)$. All field and receiver normalizations must transform together if these are reparameterized.

48. Precision, memory and the next physical interface

For original actual-clock bounds ℓ and u, the declared full scaling predicts accumulated phases in $[\ell T, uT]$ and $[n\ell T, nuT]$. For $n = \sqrt{1-a}$, their independent enclosure bands are disjoint whenever

$$a > 1 - \left(\frac{\ell}{u}\right)^2.$$

That independent-band comparison has a best-lane threshold about 2.84×10^{-9}, above the Earth example's 1.39×10^{-9} lift.

Keeping the common baseline path gives a sharper mathematical result. The full scaling implies $\theta_a(T) = \theta_0(nT)$, so

$$\Delta\theta = -\int_{nT}^{T} \nu_0(s)\,ds,$$

$$-u(1-n)T \le \Delta\theta \le -\ell(1-n)T < 0.$$

For $n = \sqrt{1-a}$, use $a/2 < 1-n \le a/(2-a)$ to obtain the entirely rational enclosure

$$-\frac{uaT}{2-a} \le \Delta\theta \le -\frac{\ell aT}{2} < 0.$$

Thus every positive lift and elapsed time has a strictly negative correlated phase change under this declared coupling, within the original window. The Earth example in lane01 gives $[-4.373897816, -4.373897806]$ nanoradians over one original bare cycle. This uses the same unwrapped branch and preparation, not independent rate fits. Fresh-copy phase readout and systematic errors must resolve the difference; no apparatus sensitivity or physical coupling is assigned. [13]

The exact elimination in sections 19–22 becomes spatial after an incidence map. A controlled response derivation must establish three-dimensional locality, isotropy, masslessness and causal speed.

Physical tasks remain: complete local energy, external V, SI calibration, energy-form response bounds and apparatus.

[13] voli_chapter3_source_field1_20261015: source, radial, energy and actual-clock derivations; exact operands, independent checks and native receipts. Previous sections 1–40 and their sources remain retained verbatim.

Data availability

The repository designated for this chapter's data and code is https://github.com/iwtbotiwtwot/sam_volume_I_ii.